

Quiz 2

1 Grading Scheme

- Question 1 is for 2 points, 1 point for each direction of the proof.
- Question 2 is for 1 point. 0.5 points for answers without clear justification.
- Question 3 is for 2 points (1 point per justification). 0.5 points for answers without justifications.

Total grade is 5.

2 Solutions

1. Proof:

($\implies$) Suppose x is an odd integer. Then $x = 2y + 1$ for some integer $y \in \mathbb{Z}$. Then we have,

$$x = 2y + 1 = (y + y) + 1 = (y + 1) + y$$

Note that $(y + 1)$ and y are consecutive integers. Hence, any odd integer is the sum of two consecutive integers.

($\impliedby$) Let y and $(y + 1)$ be two consecutive integers. Then,

$$(y + 1) + y = (y + y) + 1 = 2y + 1$$

Because $2y + 1$ is odd, we have that the sum of any two consecutive integers is an odd integer.

2. Disproof: ($\nRightarrow$)

Let $x = 0$. Then $x + 1 = 0 + 1 = 1$ is a positive integer, but $x = 0$ is not a positive integer.

3. Disproof:

Suppose $a = 5$ and $b = -5$. Let $c = -1$. Then,

$$a \times c = 5 \times (-1) = (-5) = b$$

$$b \times c = (-5) \times (-1) = 5 = a$$

Hence, $a \mid b$ and $b \mid a$. However, $a \neq b$ as $5 \neq -5$.

3 Common Errors

- Trying to prove for specific integers in Question 1.
- Not proving the converse direction in Question 1.
- Instead of consecutive integers, trying to prove the Question 1 for arbitrary integers.
- Using $x = -1$ in the disproof in Question 2.
- Switching positive integers with even integers in Question 2.
- Not justifying $a \mid b$ and $b \mid a$ in Question 3.
- Not using Def 3.2 to show $a \mid b$ in Question 3.